



EXPERIMENT TO DETERMINE THE YOUNG MODULUS OF A METALLIC ROD

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AIM

To determine Young's modulus of a metallic beam



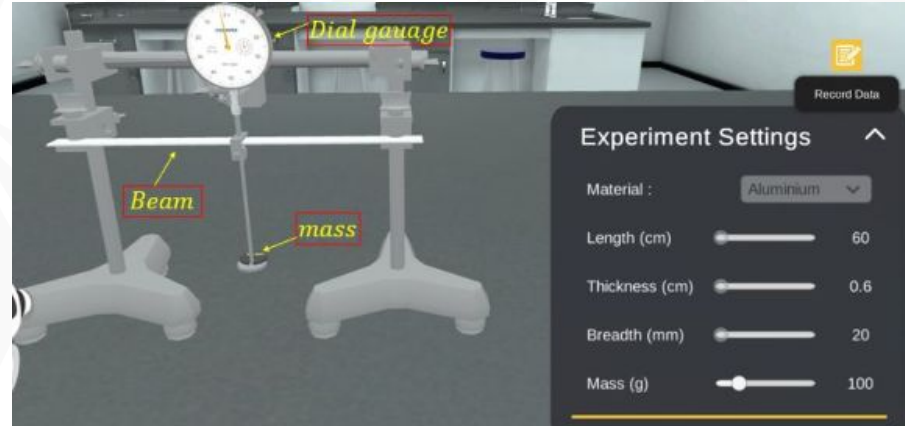
OBJECTIVES

- Define stress, strain, their types, and measuring units.
- Differentiate between types of stresses and strains.
- State units of stress and strains in various systems.
- Understand Young's modulus as a material characteristic.
- Utilize measuring gauges effectively.



APPARATUS

- Dial gauge
- Beams made from different materials
- Set of masses
- 2 mounts.





THEORY

- Young's Modulus is also known as the modulus of elasticity.
- It is a fundamental mechanical property of materials that describes their stiffness and ability to deform under an applied load.
- It is denoted by the symbol "E" or "Y" and is measured in Pascals (Pa) or N/m^2 .



THEORY

- Young's Modulus is a direct outcome of Hooke's law of elasticity.
- Young's Modulus was named after Thomas Young in the 19th century.
- It was initially developed in the 18th century by Leonhard Euler, who referred to it as "the absolute elasticity."
- Young's Modulus, E is given as:

$$E = \text{Stress/Strain}$$



THEORY

Significance of Young's Modulus Understanding Material Properties:

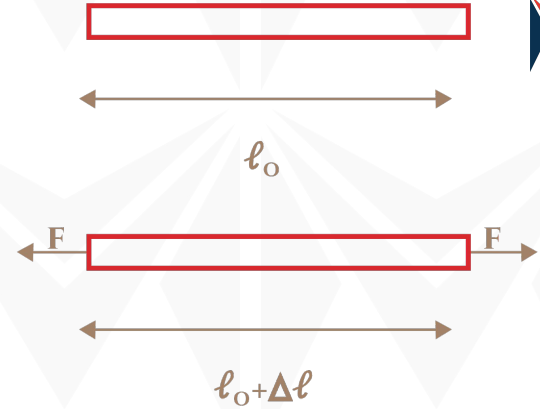
- **Stiffness:** Young's Modulus quantifies how resistant a material is to deformation when subjected to an external force.
- **Material Selection:** Engineers and material scientists use Young's Modulus to select appropriate materials for specific applications.
- **Elastic Behaviour:** Young's Modulus helps predict the elastic behaviour of materials



THEORY

- When a perpendicular force F acts on a body, it causes the object to either elongate or shorten, depending on the direction of the force.
- The stress applied to the object in this scenario is referred to as tensile stress (σ) and can be defined as the normal force per unit area.

$$\sigma = \frac{F}{A}$$



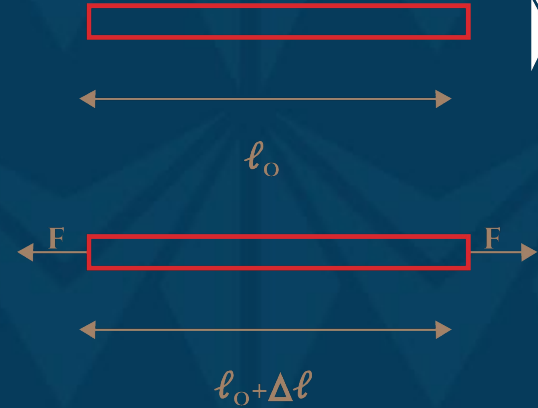


THEORY

- The relative change in dimensions for the object corresponds to the strain, denoted as ϵ .
- In the case of tensile stress, it leads to tensile strain, which is defined as :

$$\epsilon = \frac{\Delta l}{l_0}$$

- Unless the proportional limit is surpassed, stress is directly proportional to the resulting strain, a principle known as Hooke's Law,
 $\sigma \propto \epsilon \implies \sigma \propto E\epsilon$
- where E is called Young's modulus.
- The unit for E is N/m or Pascals (Pa)



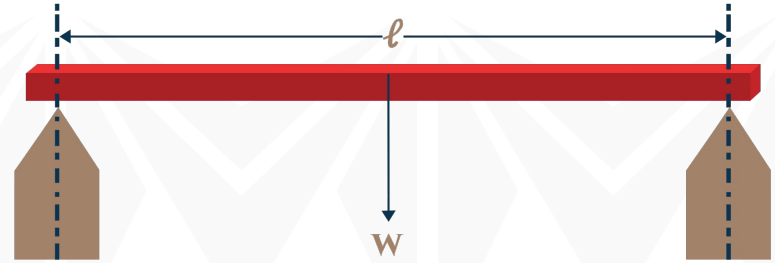


THEORY

- In this experiment, a beam of length ℓ , fixed at both ends with a load of mass m , is hung at its centre and will suffer from a depression Y under the effect of bending moment, which can be visualized as two opposite cantilevers.
- The maximum depression at the centre of the beam can be found from:

$$Y = \frac{W \ell^3}{48 E I} = \frac{m g \ell^3}{48 E I}$$

- where I is the moment of inertia of the beam and $g = 9.8 \text{ m/s}$.



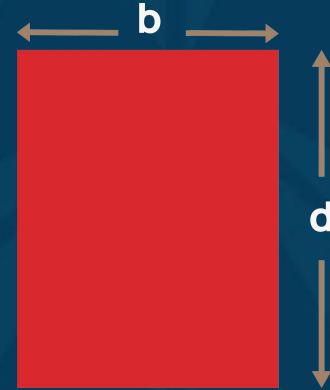


THEORY

- For a beam of rectangular cross-section, the moment of inertia, $I = (b \cdot d^3 / 12)$ where b is the breadth (width) of the rectangle and d is the thickness (depth) of the rectangle.
- Replace, I in the previous equation gives:

$$Y = \frac{g \ell^3}{4 E b d^3} m$$

- The equation is a linear equation.

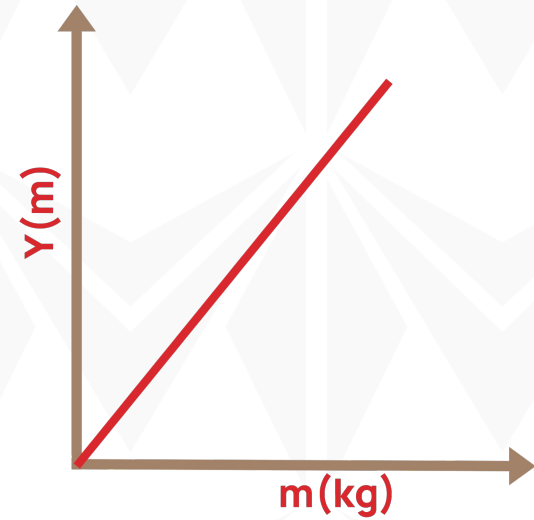




THEORY

If we plot a relation between the depression, Y versus the mass m , we get a straight line passing through the origin with a slope

$$\text{Slope} = \frac{g \ell^3}{4 E b d^3}$$





MID-LESSON QUESTIONS

Question 1:

What is the primary purpose of conducting a Young's Modulus experiment, and how does it help in material characterization?

Question 2:

In the context of Young's Modulus, what is the difference between stress and strain, and why are they crucial measurements in this experiment?



MID-LESSON ANSWERS

Answer 1:

The primary purpose of a Young's Modulus experiment is to determine how a material responds to applied forces, helping characterize its stiffness and mechanical properties.

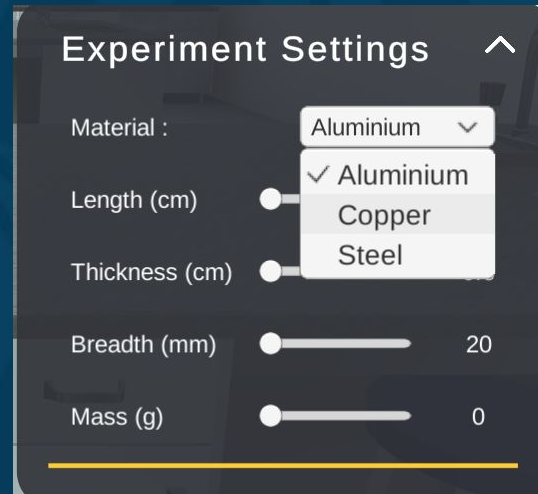
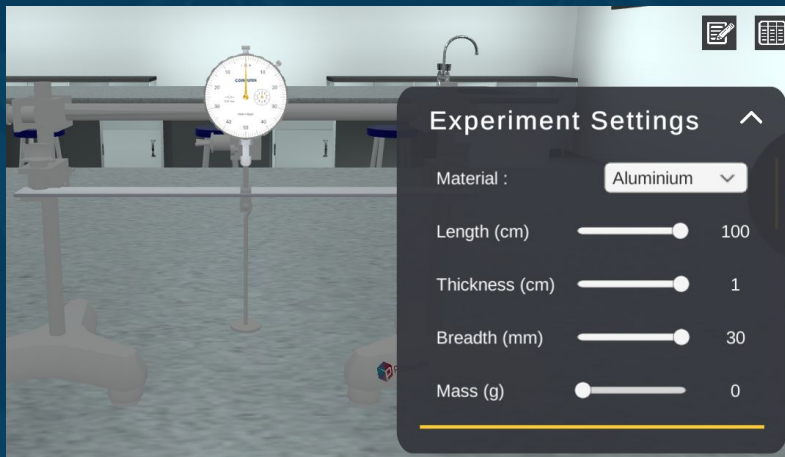
Answer 2:

In Young's Modulus, stress measures the force applied per unit area, while strain quantifies the resulting deformation. Both are crucial because they allow us to analyse the materials' behaviour under stress, helping us calculate Young's Modulus and understand its mechanical properties.



PROCEDURE

1. Select the material (e.g. Aluminium) of the beam from the drop list in the UI.
2. Set the values of length, l (e.g. 60, 70, ... 100cm), thickness/depth, d (e.g. 0.6, 0.7 ... 1.0cm), and breadth, b (20, ..., 30mm)





PROCEDURE

3. Start loading the beam by selecting the value of mass (say 50 g).
4. Click the Record button to record the value of the mass m and the corresponding depression Y .
5. Increase the mass to 100 g and Record the corresponding depression.
6. Repeat the previous steps several times.





PROCEDURE

7. The experiment has been completed and an Excel sheet that contains your recorded data will be downloaded.

Young's modulus for metallic rod

Length	100.00 cm
Thickness	1.00 cm
Breadth	30.00 mm

Material chosen: Aluminium

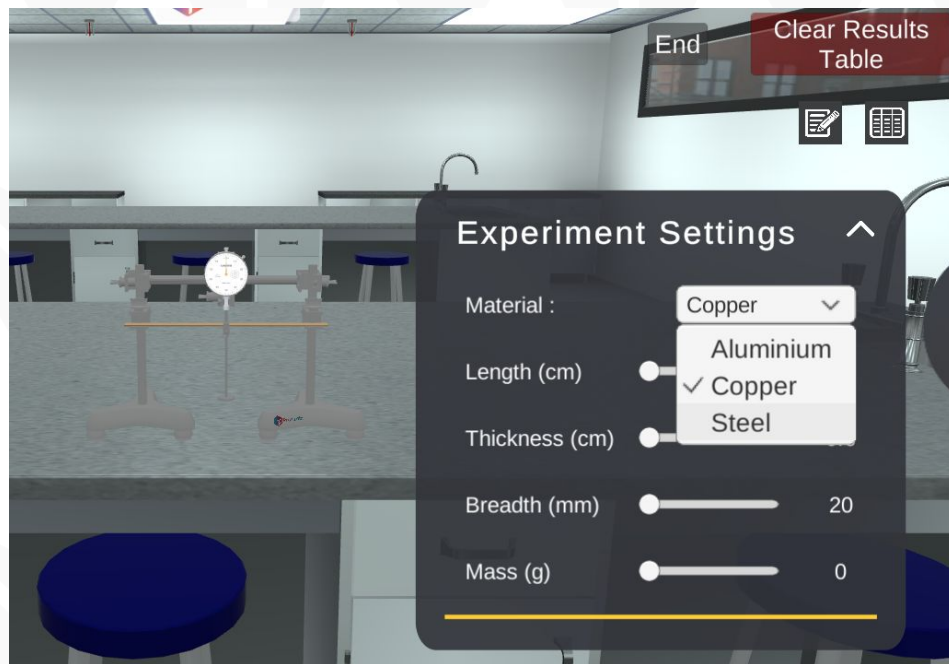
m (g)	Y(mm)
50	0.06
100	0.12
200	0.24
300	0.36
400	0.47
500	0.59





PROCEDURE

8. Repeat the entire process by changing the material to another material (say Copper).





DATA ANALYSIS

1. Convert all the values MKS system.
2. Plot a graph of $Y(m)$ vs $m(kg)$
3. Calculate the slope of the graph.
4. Use the slope together with the values of the length (ℓ), breadth (b), depth (d) to calculate Young's modulus of the material for the selected beam using:

$$E = \frac{g \ell^3}{4 b d^3 (\text{Slope})}$$

Young's modulus for metallic rod

Length	100.00 cm
Thickness	1.00 cm
Breadth	30.00 mm

Material chosen: Aluminium

	$m (g)$	$Y(mm)$
	50	0.06
	100	0.12
	200	0.24
	300	0.36
	400	0.47
	500	0.59





WALK THROUGH VIDEO





FURTHER READING

Bueche F. and Hecht E. Schaum's Easy Outlines - College Physics Crash Course. 2000 McGraw-Hill, USA.

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THANK YOU